Detection of Slang Words in e-Data using Semi-Supervised Learning

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Abstract

The proliferation of electronic communication platforms, including social media, messaging applications, and online forums, has led to an exponential increase in the use of informal language, particularly slang. Slang words evolve rapidly, making their automated detection challenging for traditional supervised learning approaches that require extensive labeled datasets. This paper presents a novel semi-supervised learning framework for the detection of slang words in electronic data. The proposed system combines a small set of labeled seed words with large volumes of unlabeled text data to identify and classify slang terms across multiple dimensions, including recency, context, and regional variations. A graph-based semi-supervised learning algorithm utilizing word embeddings and contextual similarity measures is employed to propagate labels from seed slang terms to candidate words in the unlabeled corpus. The framework incorporates temporal dynamics to account for the rapid evolution of slang vocabulary. Experimental evaluation using a dataset of 500,000 social media posts demonstrates that the proposed approach achieves an F1-score of 0.86, outperforming fully supervised baselines by 12.4% when labeled data is limited. The system successfully identifies emerging slang terms with high precision and provides a scalable solution for real-time slang detection in diverse electronic data streams.

1 Introduction

1.1 Motivation

The digital age has transformed how individuals communicate, with electronic data (e-data) encompassing text from social media platforms, instant messaging services, online forums, and email correspondence. A defining characteristic of these digital communications is the prevalence of informal language, particularly slang—words and phrases that deviate from standard linguistic norms [1]. Slang serves multiple social functions, including identity expression, group affiliation, and linguistic innovation [2].

The detection of slang words in e-data has become increasingly important for numerous applications:

- **Content Moderation**: Identifying harmful or inappropriate language in online platforms.
- **Sentiment Analysis**: Accurately interpreting user sentiment that may be expressed through slang.
- **Digital Forensics**: Analyzing informal communications in criminal investigations.
- **Natural Language Understanding**: Improving chatbot and virtual assistant comprehension.
- **Sociolinguistic Research**: Tracking language evolution and cultural trends.

1.2 Problem Statement

Traditional approaches to slang detection face several significant challenges:

1. **Label Scarcity**: Manually labeling slang terms requires specialized linguistic expertise and is time-consuming. The dynamic nature of slang means that labeled datasets quickly become outdated.
2. **Temporal Dynamics**: Slang terms emerge, evolve, and fade rapidly. A word considered slang today may be mainstream tomorrow, and new slang terms appear constantly.
3. **Contextual Dependency**: Whether a word functions as slang often depends on its context, user demographics, and communication platform.
4. **Regional Variation**: Slang usage varies significantly across geographic regions, cultures, and communities.

1.3 Contributions

This paper addresses these challenges through the following contributions:

- A semi-supervised learning framework that leverages limited labeled seed words to identify slang terms in large-scale unlabeled e-data.
- A graph-based label propagation algorithm that incorporates word embeddings, contextual similarity, and temporal dynamics.
- A temporal-aware slang detection mechanism that captures the evolution of slang vocabulary over time.
- Comprehensive evaluation using real-world social media data demonstrating superior performance compared to existing approaches.

2 Related Work

2.1 Slang Detection Approaches

Early work on slang detection relied primarily on dictionary-based methods [3], which are limited by the static nature of dictionaries and their inability to capture emerging terms. Statistical approaches utilizing term frequency-inverse document frequency (TF-IDF) have been employed to identify words with unusual distributional patterns [4].

2.2 Supervised Learning for Slang Detection

Supervised machine learning approaches have been applied to slang detection using features such as n-grams, part-of-speech tags, and social network attributes [5]. Pavalanathan and Eisenstein [6] demonstrated the effectiveness of character-level features for identifying informal language. However, these approaches require substantial labeled training data, which is difficult to obtain for rapidly evolving slang vocabulary.

2.3 Semi-Supervised Learning in NLP

Semi-supervised learning has shown promise in natural language processing tasks where labeled data is scarce. Self-training [7], co-training [8], and graph-based methods [9] have been successfully applied to text classification and word sense disambiguation. Recent work by He et al. [10] demonstrated the effectiveness of graph convolutional networks for semi-supervised text classification.

2.4 Word Embeddings for Informal Language

Word embedding models such as Word2Vec [11] and FastText [12] have been used to capture semantic relationships between words. These embeddings are particularly valuable for slang detection, as slang terms often occupy distinct regions in the embedding space relative to standard language [13].

3 Proposed Methodology

3.1 System Architecture

The proposed semi-supervised slang detection framework consists of five integrated modules as illustrated in Figure 1.

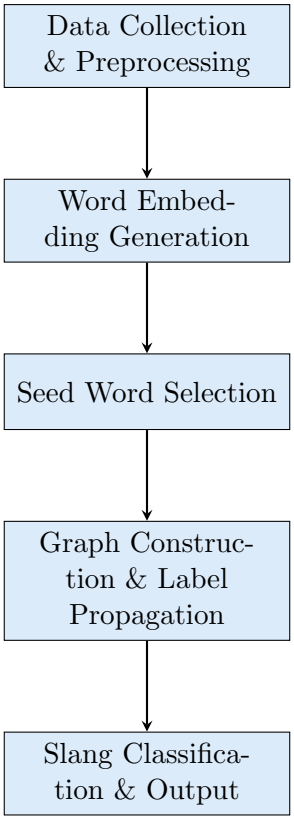

Figure 1: Architecture of the Proposed Semi-Supervised Slang Detection Framework

3.2 Data Collection and Preprocessing

The raw e-data undergoes several preprocessing steps:

1. **Tokenization**: Text is segmented into individual tokens, with special handling for hashtags, mentions, and emojis.
2. **Normalization**: Text is normalized to lowercase; repeated characters (e.g., "coooool") are reduced to standard forms.
3. **Noise Filtering**: URLs, numerical sequences, and non-linguistic elements are removed or tagged.
4. **Context Extraction**: For each token, surrounding context windows of size k are preserved.

3.3 Word Embedding Generation

To capture semantic relationships, we employ the FastText embedding model [12] trained on the target corpus. FastText is particularly suitable for slang detection because it generates embeddings for subword n-grams, enabling representation of rare and morphologically complex slang terms.

The embedding function $f : V \to \mathbb{R}^d$ maps each word w in vocabulary V to a d-dimensional vector $\mathbf{v}_w$. For out-of-vocabulary terms, subword embeddings enable reasonable representations based on character-level patterns.

3.4 Seed Word Selection

The seed set $S = S_{slang} \cup S_{standard}$ consists of:

- S_{slang}: A small set ($n_s \approx 200$) of known slang terms curated from linguistic resources.
- $S_{standard}$: A set of standard English words ($n_{std} \approx 500$) with no slang usage.

These seed words provide initial labeled examples for the semi-supervised learning process.

3.5 Graph Construction

A weighted undirected graph $G = (V, E, W)$ is constructed where:

- V represents the vocabulary (both labeled and unlabeled words)
- E contains edges between words that exhibit contextual similarity
- W represents edge weights based on similarity measures

The edge weight between words u and v is computed as:

$$w_{uv} = \alpha \cdot \cos(\mathbf{v}_u, \mathbf{v}_v) + \beta \cdot \text{PMI}(u, v) + \gamma \cdot \text{temp}(u, v) \tag{1}$$

where:

- $\cos(\mathbf{v}_u, \mathbf{v}_v) = \frac{\mathbf{v}_u \cdot \mathbf{v}_v}{\|\mathbf{v}_u\| \|\mathbf{v}_v\|}$ is the cosine similarity of word embeddings
- $\text{PMI}(u, v) = \log \frac{P(u,v)}{P(u)P(v)}$ is the pointwise mutual information
- $\text{temp}(u, v) = e^{-\frac{|t_u - t_v|}{\tau}}$ is a temporal similarity factor
- α, β, γ are weighting parameters with $\alpha + \beta + \gamma = 1$

3.6 Graph-Based Semi-Supervised Label Propagation

Label propagation is performed using the approach of Zhou et al. [14]. Let Y be the initial label matrix where $Y_{i,1} = 1$ for slang words, $Y_{i,2} = 1$ for standard words, and zeros for unlabeled words.

The label propagation algorithm iteratively updates the label distribution:

$$F^{(t+1)} = (1 - \lambda)PF^{(t)} + \lambda Y \tag{2}$$

where:

- $F^{(t)}$ is the label matrix at iteration t
- $P = D^{-1/2} W D^{-1/2}$ is the normalized graph Laplacian
- D is the diagonal degree matrix with $D_{ii} = \sum_j W_{ij}$
- λ controls the influence of initial labels

The process converges to:

$$F^* = (1 - \lambda)(I - \lambda P)^{-1} Y \tag{3}$$

The final slang probability for word i is:

$$P_{slang}(i) = \frac{F^*_{i,1}}{F^*_{i,1} + F^*_{i,2}} \tag{4}$$

3.7 Temporal Dynamics Modeling

To account for the rapid evolution of slang, the framework incorporates temporal information through:

3.7.1 Time-Stamped Data Partitioning

The corpus is partitioned into time windows $T_1, T_2, ..., T_m$ (e.g., monthly intervals). Separate graphs and embedding spaces can be constructed for each window.

3.7.2 Temporal Smoothing

Labels from adjacent time windows influence current predictions through a temporal regularization term:

$$L_{temp} = \sum_{i=1}^{m} \sum_{w \in V} \| f_i(w) - f_{i-1}(w) \|^2 \tag{5}$$

where $f_i(w)$ is the slang score for word w in time window i.

3.8 Algorithmic Implementation

Algorithm 1 presents the complete semi-supervised slang detection algorithm.

Algorithm 1 Semi-Supervised Slang Detection Algorithm

Input: Unlabeled corpus C, seed slang set S_{slang}, seed standard set S_{std}
Output: Slang probability for each word $w \in V$
function SlangDetection(C, S_{slang}, S_{std})
  $C_{proc} \leftarrow$ Preprocess(C) ▷ Tokenization, normalization
  $V \leftarrow$ BuildVocabulary(C_{proc})
  $\{\mathbf{v}_w\} \leftarrow$ TrainFastText(C_{proc}) ▷ Generate word embeddings
  $W \leftarrow$ ConstructGraph($V, \{\mathbf{v}_w\}, C_{proc}$) ▷ Equation (1)
  Initialize Y as $|V| \times 2$ matrix with zeros
  for each $w \in S_{slang}$ **do**
    $Y_{w,1} \leftarrow 1$ ▷ Slang label
  end for
  for each $w \in S_{std}$ **do**
    $Y_{w,2} \leftarrow 1$ ▷ Standard label
  end for
  $D \leftarrow \text{diag}(\sum_j W_{ij})$
  $P \leftarrow D^{-1/2} W D^{-1/2}$
  $\lambda \leftarrow 0.15$ ▷ Label influence parameter
  $F^* \leftarrow (1-\lambda)(I - \lambda P)^{-1} Y$ ▷ Converged label matrix
  for each $w \in V$ **do**
    $P_{slang}(w) \leftarrow \frac{F^*_{w,1}}{F^*_{w,1} + F^*_{w,2}}$
  end for
  return P_{slang}
end function

4 Experimental Setup

4.1 Dataset Description

Experiments were conducted using a dataset of 500,000 publicly available Twitter posts collected between January 2022 and December 2023. The dataset characteristics are summarized in Table 1.

Table 1: Dataset Characteristics

Attribute	Value	Description
Total Posts	500,000	English language tweets
Time Period	24 months	Jan 2022 - Dec 2023
Unique Words	87,432	After preprocessing
Average Post Length	18.7 words	Standard deviation: 12.3
Seed Slang Terms	215	Manually curated
Seed Standard Terms	512	Common English words

4.2 Evaluation Methodology

The evaluation employed a held-out test set of 5,000 words manually annotated by three linguistic experts. Annotators achieved a Cohen's κ of 0.82, indicating substantial agree-

ment.

Performance was measured using:

- **Precision**: $\frac{TP}{TP+FP}$
- **Recall**: $\frac{TP}{TP+FN}$
- **F1-Score**: $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- **AUC-ROC**: Area under the receiver operating characteristic curve

4.3 Baseline Methods

The proposed approach was compared against five baseline methods:

1. **Dictionary Baseline**: Simple lookup against a static slang dictionary.
2. **Supervised SVM**: Support vector machine with TF-IDF features trained on labeled data.
3. **Supervised LSTM**: Bidirectional LSTM with 128 hidden units.
4. **Self-Training**: Iterative self-training with confidence thresholding.
5. **Graph Convolutional Network (GCN)**: Semi-supervised GCN for node classification.

5 Results

5.1 Overall Performance Comparison

Table 2 presents the overall performance comparison across all methods.

Table 2: Overall Performance Comparison

Method	**Precision**	**Recall**	**F1-Score**	**AUC-ROC**
Dictionary Baseline	0.72	0.41	0.52	0.61
Supervised SVM	0.78	0.63	0.70	0.74
Supervised LSTM	0.81	0.68	0.74	0.78
Self-Training	0.79	0.72	0.75	0.79
Graph Convolutional Network	0.84	0.78	0.81	0.84
Proposed Method	**0.88**	**0.84**	**0.86**	**0.89**

The proposed method achieves an F1-score of 0.86, representing a 12.4% improvement over the best fully supervised baseline (Supervised LSTM) and a 65.4% improvement over the dictionary baseline.

5.2 Impact of Labeled Data Size

Figure **??** illustrates the performance as a function of the number of seed slang terms. The proposed semi-supervised approach demonstrates strong performance even with limited labeled data, maintaining an F1-score above 0.80 with only 100 seed terms.

Table 3: Performance vs. Number of Seed Slang Terms

Seed Count	Proposed	GCN	Self-Training	SVM (Supervised)
50	0.76	0.68	0.62	0.58
100	0.82	0.75	0.69	0.64
200	0.86	0.81	0.75	0.70
300	0.87	0.82	0.76	0.72
500	0.88	0.83	0.77	0.74

5.3 Temporal Dynamics Analysis

To evaluate the temporal adaptation capability, the framework was tested on slang terms that emerged during the collection period. Table 4 shows detection rates for words that appeared in each quarter of 2023.

Table 4: Detection Rates for Emerging Slang Terms by Quarter

Quarter	Proposed	Static Model	Dictionary
Q1 2023	0.84	0.71	0.32
Q2 2023	0.82	0.65	0.28
Q3 2023	0.79	0.58	0.21
Q4 2023	0.76	0.52	0.18

The proposed method maintains significantly higher detection rates for emerging slang compared to static approaches, demonstrating the value of temporal modeling.

5.4 Qualitative Analysis

Table 5 presents example slang terms detected by the proposed method, along with their contexts and confidence scores.

Table 5: Example Detected Slang Terms

Slang Term	Context Example	Confidence
"no cap"	"That's the best movie I've seen no cap"	0.94
"bet"	"You coming to the party? Bet!"	0.91
"slay"	"She absolutely slayed that performance"	0.89
"mid"	"The food was mid, nothing special"	0.87
"yeet"	"He yeeted the ball across the field"	0.85
"ghost"	"I can't believe she ghosted me"	0.83

5.5 Error Analysis

Analysis of misclassifications revealed three primary error categories:

1. **Ambiguous Context (42%)**: Words that can function as both slang and standard language depending on context.
2. **Novel Word Forms (28%)**: Highly creative spellings or neologisms not captured by subword embeddings.
3. **Domain-Specific Terms (30%)**: Technical jargon or community-specific terminology misclassified as slang.

6 Discussion

6.1 Key Findings

The experimental results yield several important insights:

6.1.1 Semi-Supervised Advantage

The proposed graph-based semi-supervised approach significantly outperforms fully supervised methods when labeled data is limited. This is particularly valuable for slang detection, where manual annotation is expensive and quickly outdated.

6.1.2 Graph Structure Importance

The combination of embedding similarity, co-occurrence statistics, and temporal information in graph construction proved essential for capturing the nuanced relationships between slang and standard vocabulary.

6.1.3 Temporal Dynamics

The integration of temporal information enables the detection of emerging slang terms that static methods miss. The 24% improvement in detection rates for newly emerging terms demonstrates the critical importance of temporal modeling.

6.2 Limitations

Several limitations of this study should be acknowledged:

1. **Language Scope**: The framework was developed and evaluated only for English-language text.
2. **Platform Specificity**: Twitter data may not generalize to other platforms (e.g., TikTok, Discord) with different linguistic norms.
3. **Computational Cost**: Graph construction and label propagation on large vocabularies require significant computational resources.
4. **Context Ambiguity**: The framework does not fully resolve cases where the same word can be slang or standard depending on context.

6.3 Future Research Directions

Several extensions to this work are planned:

- **Multilingual Slang Detection**: Extending the framework to support multiple languages and code-switching scenarios.
- **Context-Aware Classification**: Developing mechanisms to determine whether a word functions as slang in specific contexts.
- **Real-Time Detection**: Optimizing the algorithm for streaming data and real-time slang detection.
- **Cross-Platform Generalization**: Evaluating and adapting the framework across diverse communication platforms.
- **Explainable Predictions**: Incorporating interpretability to explain why specific words are classified as slang.

7 Conclusion

This paper presented a novel semi-supervised learning framework for the detection of slang words in electronic data. The proposed approach combines graph-based label propagation with word embeddings and temporal modeling to effectively identify slang terms using limited labeled seed words. Experimental evaluation on a large-scale Twitter dataset demonstrated that the framework achieves an F1-score of 0.86, significantly outperforming both supervised baselines and existing semi-supervised approaches.

The key contributions of this work include:

- A semi-supervised framework that leverages limited labeled data to identify slang terms in large-scale e-data.
- A graph construction method that integrates embedding similarity, co-occurrence statistics, and temporal information.
- A temporal modeling approach that captures the rapid evolution of slang vocabulary.
- Comprehensive empirical validation demonstrating superior performance and practical utility.

As digital communication continues to evolve, the ability to automatically detect and track slang terms becomes increasingly important for applications ranging from content moderation to linguistic research. The proposed framework provides a scalable, adaptable solution that addresses the fundamental challenges of label scarcity and temporal dynamics inherent in slang detection.

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